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Standardised Operation Procedure

M2 Repair for Global Repair Centers

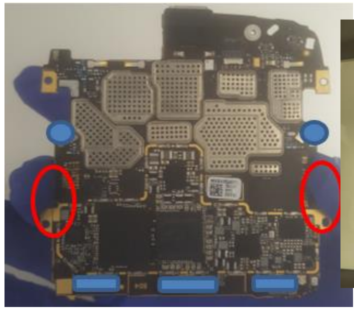
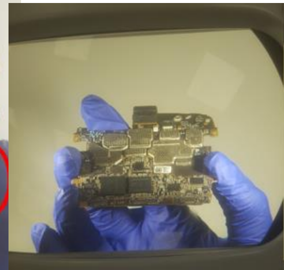
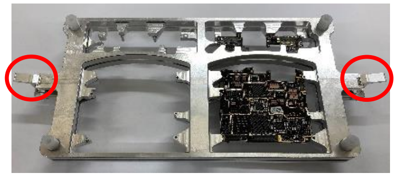
SOP Unique No.	M2RWI010	**Document Version**	**3**
Originator (s)	H. Willis, P. Grudnik	**Document released**	**27/01/2020**
Department	P2i Technology Department	**Document Updated**	**29/07/2020**
Process Description	M2 Repair (Repair Centers)	**Document Updated to be Generic**	**25/08/2020**
Operator level required:	Qualified Repair Operator		
Training Requirements:	Ensure operator has received M2 repair training		
Approved by	A. McLeod		

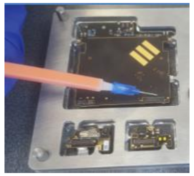
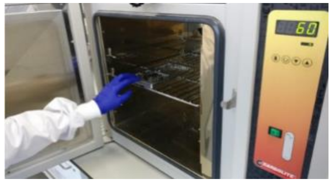
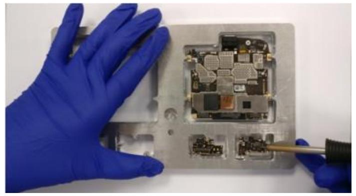
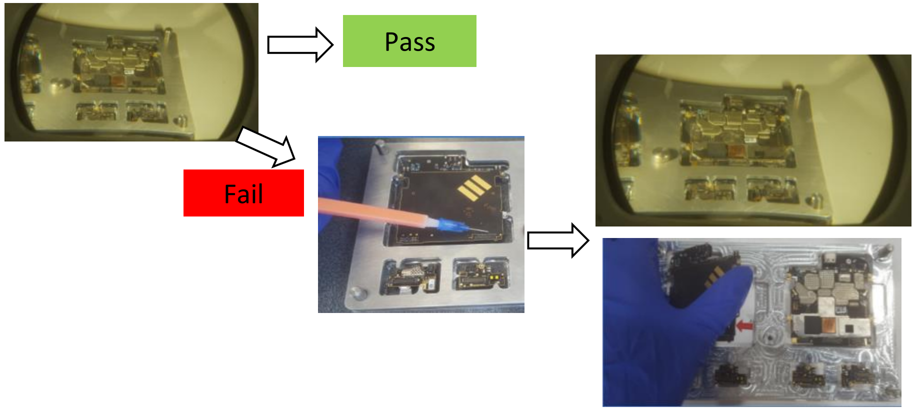
Revision	Date	Reason	Approved by
V1	27/01/2020	Initial Document	Alex McLeod
V2	29/07/2020	Updated to be generic	Alex McLeod
V3	25/08/2020	Updated to cover new jigs	Alex McLeod

Process Description	M2 Repair				**Health and Safety / PPE**
Element Description	Visual Quality Check	**P2i Confidential - For use by Motorola Repair Centers Only**			Prior to commencement please read relevant risk assessment for Hazards and Risks identified
Operator level required:	Level 2-Intermediate				

	Safety key point		Critical point	Q	Quality check
P	Poka Yoke	3	Standard process		Knack point

Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

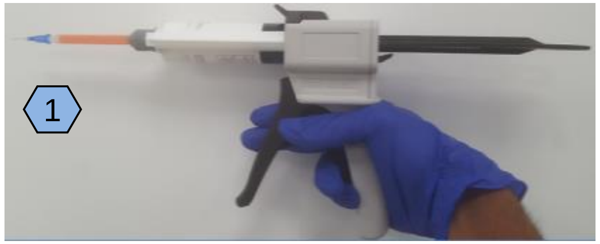
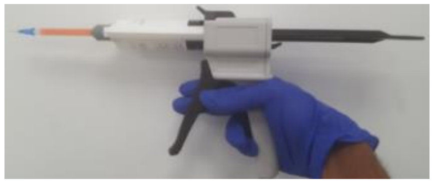
	Major Step	Key	Operation detail	Explanation/Examples/Diagrams
1	**Prepare area**		A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. Linear Spudger ESD safe 2. Magnifying Lamp 3. Double Re-work Jig 4. PCBA Specific Dunkable Pass-Fail Criteria for product being repaired **C. Ensure ESD strap is connected and worn at all time.** **D. Only authorized or trained personnel to undertake this task.**	1  2  3   4
2	**Primary Inspection**	Q	A. Turn on magnifying lamp. B. Hold the PCBAs without touching any M2. C. Hold PCBAs under the lamp and compare with **'Dunkable Pass-Fail Criteria for Lenovo Production Line'** for specific PCBA being reworked	  
3	**Place in Re-work Jig**	Q	A. Place boards to be re-worked in the designated sections of the product specific re-work jig, assemble the two halves and tighten the screw/clip together.	  

4	**Re-work**		A. Re-work the failed positions.	  
5	**Secondary Inspection**		A. Once the boards have been reworked, the magnifying lens can be used to inspect the reworked points according to the pass/fail criteria. B. Any failed points will need to undergo rework and inspection again. C. Upon passing the rework boards, the rework jig can be disassembled, and the boards sent to board testing. D. PCBAs complete device will need to be reassembled and sent to post assembly CQA testing.	

Process Description	M2 Repair		**Health and Safety / PPE**
Element Description	Storing Cartridges	**P2i Confidential - For use by Motorola Repair Centers Only**	**Prior to commencement please read relevant risk assessment for Hazards and Risks identified**
Operator level required:	Level 2-Intermediate		

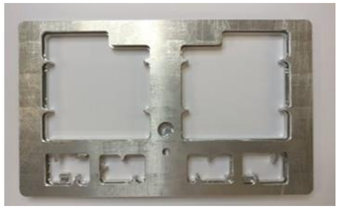
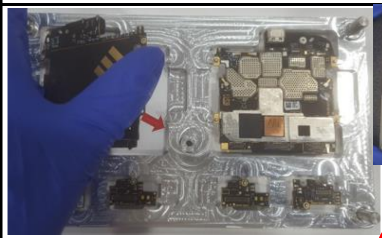
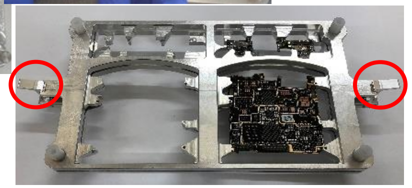
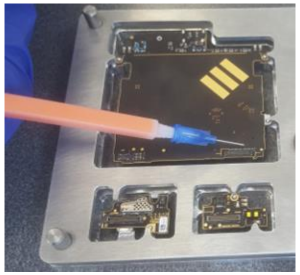
Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

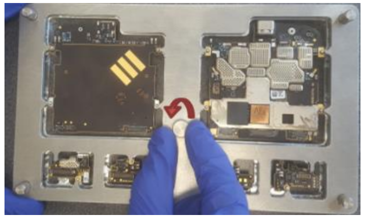
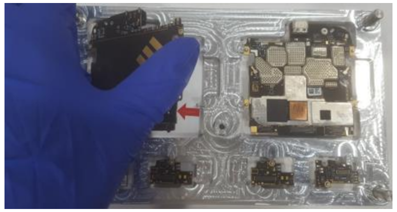
	Safety key point		Critical point	O	Quality check
P	Poka Yoke	3	Standard process		Knack point

	Major Step	**Key**	**Operation detail**	**Explanation/Examples/Diagrams**
1	Prepare area		A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. M2 Dispensing Gun 2. Blue Roll **C. Only authorized or trained personnel to undertake this task.** D. P2i recommend disposing of the mixing tip and nozzle at the end of the shift. If shifts run 24 hours a day, P2i recommend changing the mixing nozzle and tip once every 24 hurs. E. If the M2 dispense mechanism seizes after time idling P2i recommend changing the mixing nozzle and tip as curing may have occured in the mixing nozzle.	 
2	**Disassembly**		A. Remove the dispensing and mixing tip from the gun and replace the cap on the cartridge. B. Dispose of the tips in the sharps bin.	  
3	**Storage**		A. Place blue roll on the cabinet shelf to prevent contamination of the shelf. B. Place the dispensing gun on the shelp as shown and securely shut the cabinet doors.	 

Process Description	M2 Repair	**Health and Safety / PPE**
Element Description	Using Re-work Jigs **P2i Confidential - For use by Motorola Repair Centers Only**	**Prior to commencement please read relevant risk assessment for Hazards and Risks identified**
Operator level required:	Level 2-Intermediate	

	Safety key point		Critical point	O	Quality check	Safety Shoes		Lab coat	✓
						Light Eye Protection		Hearing Protection	✓
P	Poka Yoke	3	Standard process		Knack point	Goggles	✓		
						Gloves - Polyflex			

	Major Step	**Key**	**Operation detail**	**Explanation/Examples/Diagrams**
1	Prepare area		A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. Rework Jig 2. M2 Dispensing Gun **C. Ensure ESD strap is connected and worn at all time.** **D. Only authorized or trained personnel to undertake this task.**	1  2 
2	Place Boards	3	A. Insert boards into designated slots. B. Connect top and bottom half of the rework jig. C. Screw or clip into place	  
3	**Dispense and Cure**	3	A. Dispense as directed in the M2 application SOP before curing as directed in the curing SOP.	

4	**Remove Boards**	3	A. Unscrew/unclip the top half of the jig and remove the boards.	 
5	**Reference Photo**	3	A. Please ensure that all PCBAs are correctly loaded into re-work jigs - choose correct jig for correct PCBA **B. If PCBAs are not loaded in the correct location PCBAs may be damaged by jig or move during re-work** **C. Please see image to the right for examples of how to load the individual boards.**	 

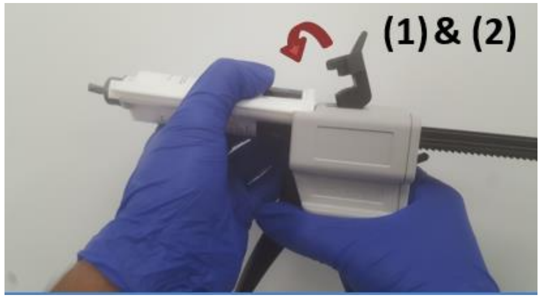
Process Description	M2 Repair	
Element Description	Loading the Cartridge	**P2i Confidential - For use by Motorola Repair Centers Only**
Operator level required:	Level 2-Intermediate	

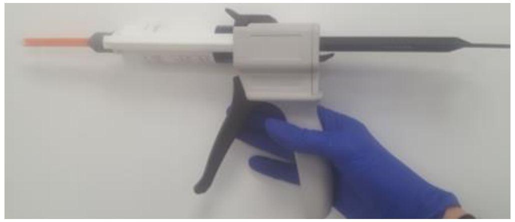
✚	Safety key point	▼	Critical point	O	Quality check
P	Poka Yoke	3	Standard process	⬭	Knack point

Health and Safety / PPE

Prior to commencement please read relevant risk assessment for Hazards and Risks identified

Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles	✓		
Gloves - Polyflex			

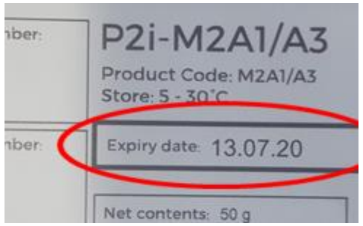
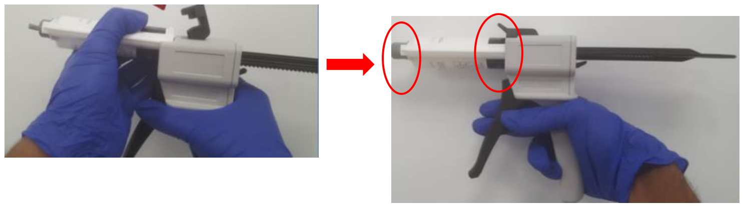
Major Step		Key	Operation detail	Explanation/Examples/Diagrams
1	Prepare area	✚	A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. Pre-filled M2 Cartridge 2. 10:1 System Dispenser (Oval) 3. Mixer-Quadro, Luer Lock Tip for 10:1 Ratio 4. Dispensing Tip, Blunt End, 1.0" Long 22ga Blue **D. Only authorized or trained personnel to undertake this task.**	1  2  3 4  
2	**Insert Cartridge**		A. Slide the cartridge (1) into the system dispenser (2). B. Once the cartridge is fully inserted, clip it into place.	

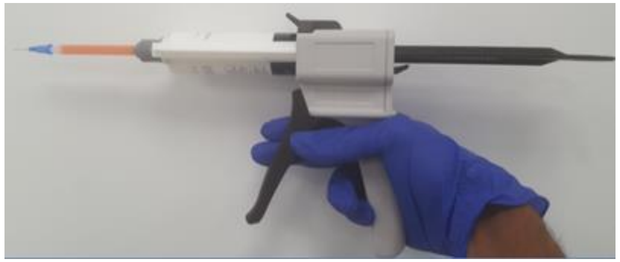
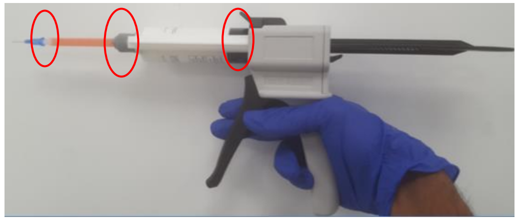
3	**Mixing Tip**	Q	A. Screw on the mixing tip (3) and purge onto blue roll. B. Ensure material has filled the mixing nozzle, and run through at least once, end-to-end, before beginning dispense with a new nozzle	 
4	**Dispensing Tip**		A. Attach the dispensing tip (4) and purge onto blue roll.	 

Process Description	M2 Repair	**Health and Safety / PPE**
Element Description	**P2i Confidential - For use by Motorola Repair Centers Only**	Prior to commencement please read relevant risk assessment for Hazards and Risks identified
Operator level required:	Level 2-Intermediate	

Symbol	Meaning	Symbol	Meaning	Symbol	Meaning
✚	Safety key point	▼	Critical point	Q	Quality check
P	Poka Yoke	3	Standard process	⬭	Knack point

PPE		PPE	
Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles	✓		
Gloves - Polyflex			

Major Step		Key	Operation detail	Explanation/Examples/Diagrams
1	Prepare area	✚	A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. Blue Roll 2. M2 Cartridge 3. M2 Dispensing Gun 4. Mixing Tip 5. Dispensing Tip **C. Only authorized or trained personnel to undertake this task.**	
2	**Check Date**	Q	A. First, check the cartridge is in date. *(If out-of-date, DO NOT USE, dispose of in designated disposal bin)*	
3	**Leak Test 1**	▼	A. Put the M2 cartridge in the gun. B. Lay out blueroll to catch any spills. C. Squeeze the trigger and check seams and the pistons for leaks.	

4	**Assemble**	Q	A. Assemble the dispensing gun as shown in the relevant SOP.	
5	**Leak Test 2**	Q	A. Hold the fully assembled dispensing gun over blue roll. B. Squeeze the trigger and check seams and the pistons for leaks.	

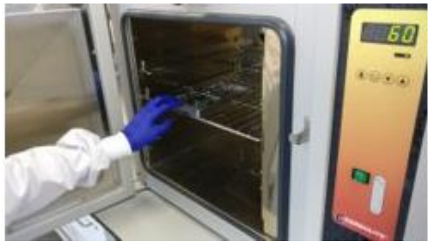
Process Description	M2 Repair	
Element Description	Curing	**P2i Confidential - For use by Motorola Repair Centers Only**
Operator level required:	Level 2-Intermediate	

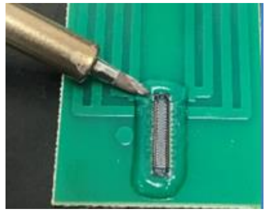
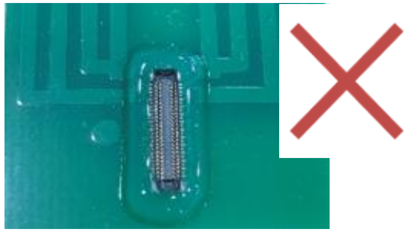
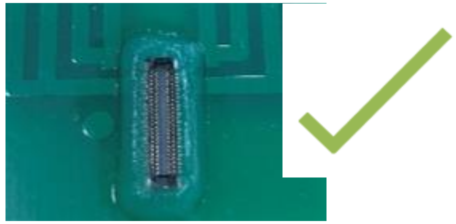
Health and Safety / PPE

Prior to commencement please read relevant risk assessment for Hazards and Risks identified

Safety Shoes		Lab coat	
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

Key		Key		Key	
✚	Safety key point	▼	Critical point	O	Quality check
P	Poka Yoke	3	Standard process	⬭	Knack point

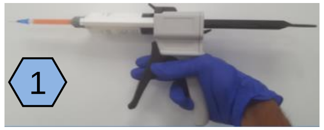
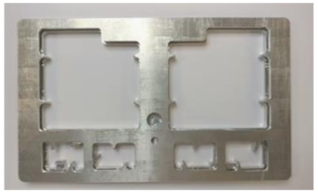
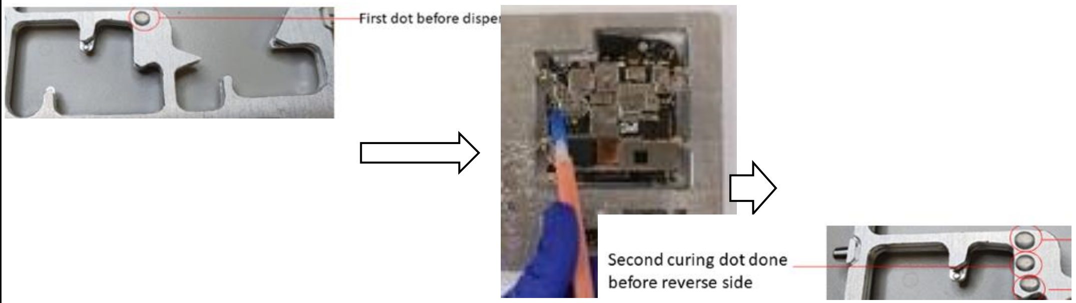
Major Step		Key	Operation detail	Explanation/Examples/Diagrams ✓
1	**Prepare area**	✚	A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. Oven set to 60 Degrees Celcius 2. M2 Soldering Iron set to 435 Degrees Celcius **C. Ensure ESD strap is connected and worn at all time.** **D. Only authorized or trained personnel to undertake this task.**	1  2 
2	**Oven Cure (Primary Cure)**	▼	A. Ensure the oven is preheated to 60°C B. Place the completed boards, still in the jig, into the oven. Close the door. C. Leave the samples for 10minutes (set a timer) before removing them from the oven. D. Primary oven cure will change the M2 from a liquid to a gel that is firm, but sticky to touch	

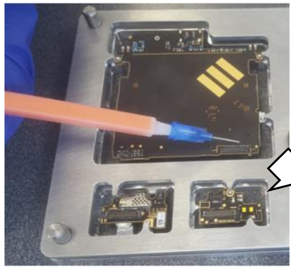
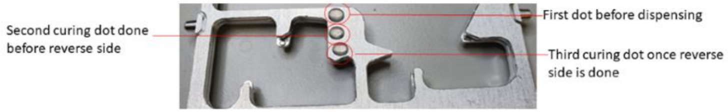
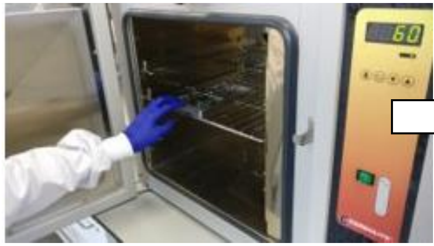
3	**Solder Cure (Secondary Cure)**		A. Heat the M2 soldering iron to 435°C. *(Use only the M2 soldering iron, ensure it is free from contamination)* B. Position it as shown and press **lightly** on the M2 for 3 seconds. Repeat this until all the M2 has been treated.	 
4	**Check**	Q	A. Check that the M2 has gone matte. If it has, the M2 has been cured. M2 should remain in place when touched after secondary cure B. M2 searing removes the tacky nature of M2, allowing handling of the M2 without it sticking to hands/tools and being removed. C. If the M2 remains attached to the board and is not removed by touching with a gloved hand, then secondary cure has been successful.	 

Process Description	M2 Repair		Health and Safety / PPE
Element Description	Applying Curing Dots	P2i Confidential - For use by Motorola Repair Centers Only	Prior to commencement please read relevant risk assessment for Hazards and Risks identified
Operator level required:	Level 2-Intermediate		

✚	Safety key point	▼	Critical point	O	Quality check
P	Poka Yoke	3	Standard process	⬭	Knack point

Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

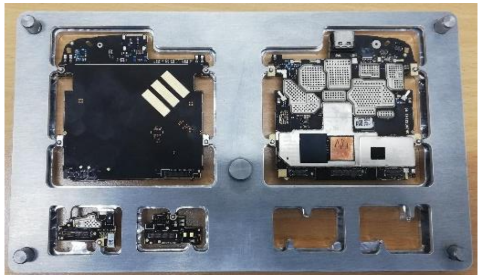
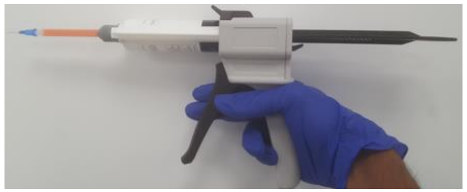
Major Step		Key	Operation detail	Explanation/Examples/Diagrams
1	Prepare area	✚	A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. M2 Dispensing Gun 2. Oven set to 60 Degrees Celcius 3. Soldering Iron set to 435 Degrees Celcius 4. Rework Jig 5. Metal Tweezers **C. Ensure ESD strap is connected and worn at all time.** **D. Only authorized or trained personnel to undertake this task.**	1   3  4  5 
2	**Dot 1 and 2**	▼	A. Before dispensing on/around any components, dispense a 3mm wide dot in the top corner of a space on the jig. B. Complete any masking on the first side. C. Dispense a second 3mm wide dot beneath the first.	

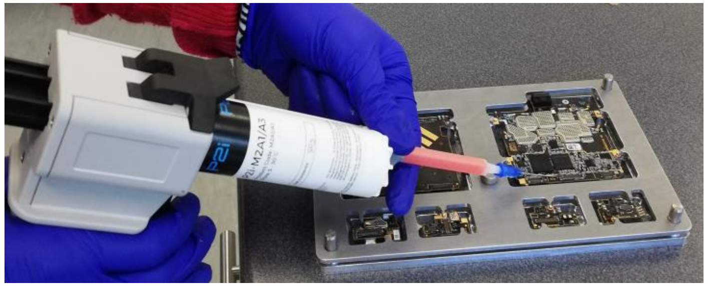
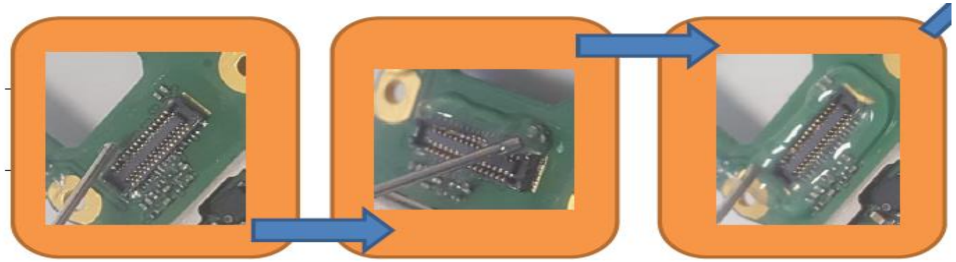
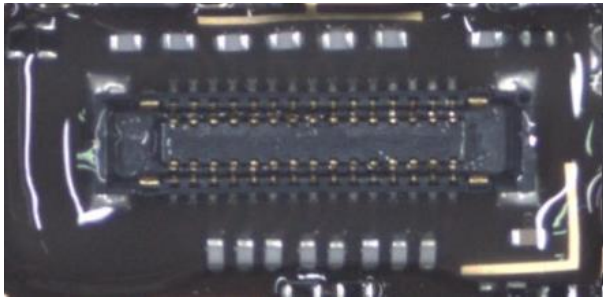
3	**Dot 3**	Q	A. Complete any masking on the second side. B. Dispense a third 3mm dot under the first two	 
4	**Cure**	Q	A. Cure according to the cure SOP.	 
5	**Curing Check**		A. Use metal tweezers to assess if the M2 is cured or not. If the M2 is still liquid then the cure has failed and any M2 must be removed from the board and re-applied with quality checked M2.	 

Process Description	M2 Application Specific	**Health and Safety / PPE**
Element Description	M2 Application on B2B connectors **P2i Confidential - For use by Motorola Repair Centers Only**	**Prior to commencement please read relevant risk assessment for Hazards and Risks identified**
Operator level required:	Level 2-Intermediate	

	Safety key point		Critical point		Quality check
	Poka Yoke	3	Standard process		Knack point

Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

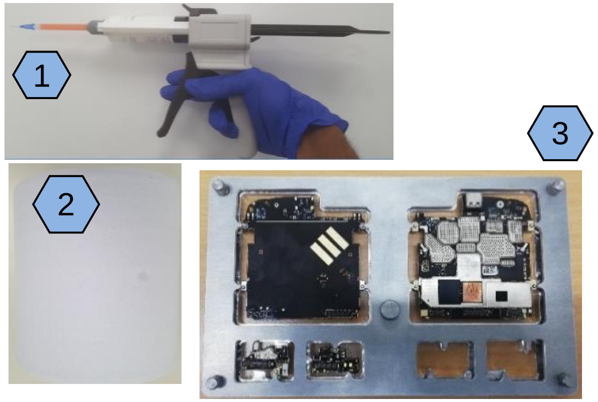
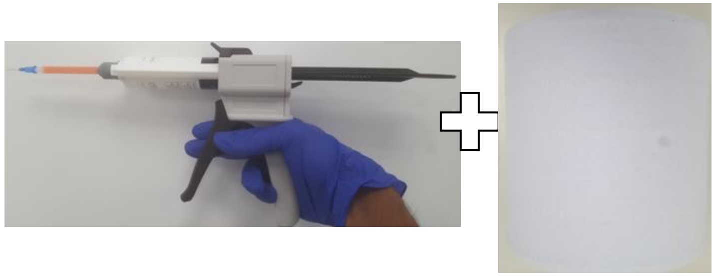
Major Step		Key	Operation detail	Explanation/Examples/Diagrams
1	**Prepare area**		A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. M2 Dispensing Gun 2. Blue Roll 3. Re-work jig with PCBAs **C. Only authorized or trained personnel to undertake this task.**	1  2  3 
2	**Set up**	3	A. Ensure that the dual syringe is clipped into place.Hold the gun with the dominant hand and the other used for support B. Ensure M2 curing check has been performed prior M2 application onto product. B. Dispense some M2 onto a paper towel to remove any air bubbles and ensure the M2 is coming out of the gun properly.	 

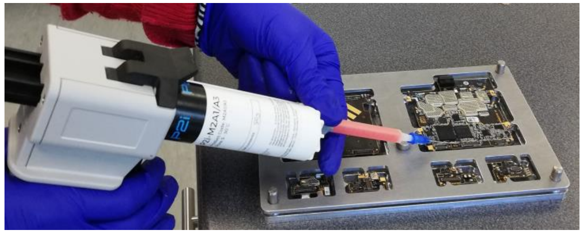
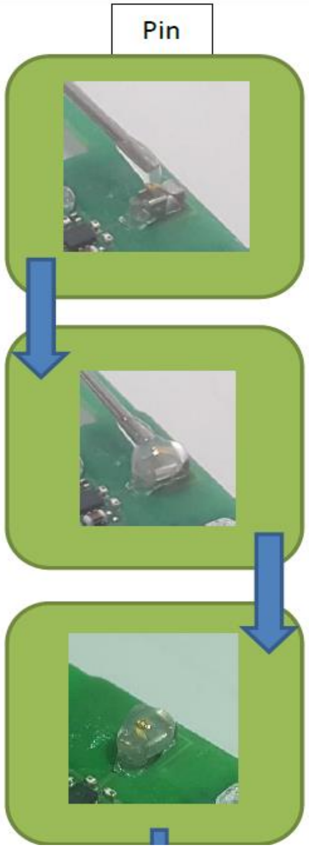
3	**Point Specfic Application**		A. M2 Ring Fence application is correct dispensing method to protect B2B connectors B. To apply, bring the grey handle towards you to dispense the material, whilst slowly moving the needle tip above the part of the PCBA which is being covered.	 
4	**VQC**	O	A. After curing perform VQC as per VQC WI and 'VQC Pass/Fail Criteria'. B. Most critial points during VQC of B2B connectors are: - **M2 Z-Height: the M2 must be higher than the B2B connector** - **There are no gaps between the B2B pins and the M2** - **M2 does not overflow into the edges of B2B or inside of B2B**	

Process Description	M2 Repair	Health and Safety / PPE
Element Description	M2 Application on universal Pin connectors	Prior to commencement please read relevant risk assessment for Hazards and Risks identified
Operator level required:	Level 2-Intermediate **P2i Confidential - For use by Motorola Repair Centers Only**	

(safety cross symbol)	Safety key point	(red triangle)	Critical point	O	Quality check
P	Poka Yoke	3	Standard process	(ellipse)	Knack point

Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

	Major Step	Key	Operation detail	Explanation/Examples/Diagrams
1	**Prepare area**	(safety cross symbol)	A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. M2 Dispensing Gun 2. Blue Roll 3. Re-work jig with PCBAs **C. Only authorized or trained personnel to undertake this task.**	
2	**Set up**	3	A. Ensure that the dual syringe is clipped into place.Hold the gun with the dominant hand and the other used for support B. Ensure M2 curing check has been performed prior M2 application onto product. B. Dispense some M2 onto a paper towel to remove any air bubbles and ensure the M2 is coming out of the gun properly.	

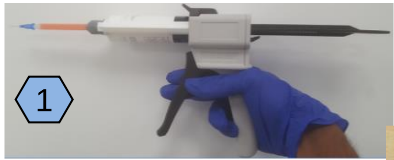
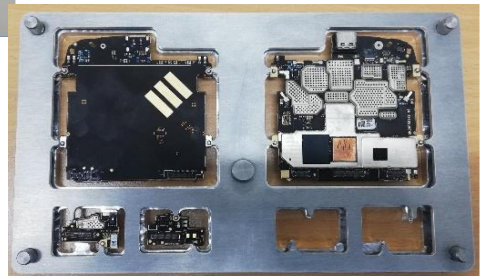
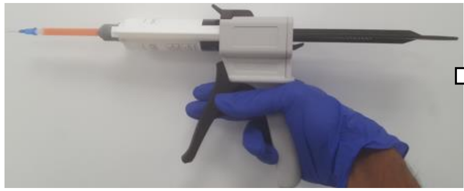
3	**Point Specfic Application**	3	A. To apply, bring the grey handle towards you to dispense the material, whilst slowly moving the needle tip from the back of the pin and apply M2 towards the gold tip, **ensure the pin is fully covered on all sides**	 

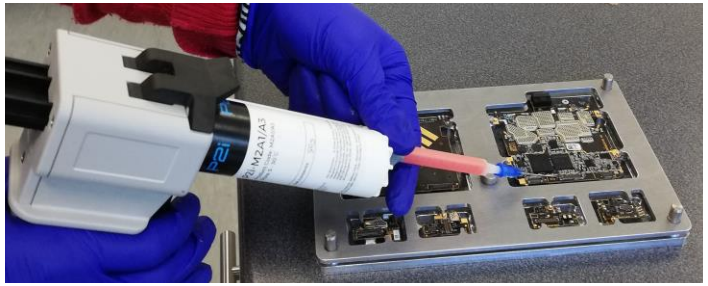
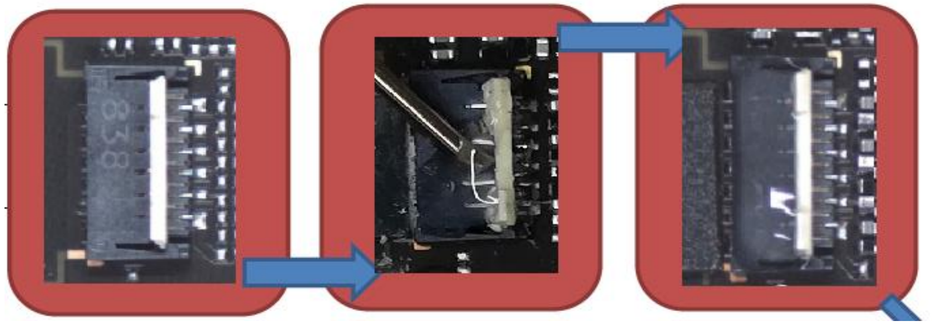
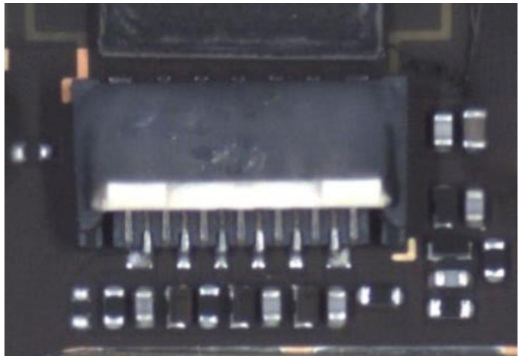
4	**VQC**	Q	A. After curing perform VQC as per VQC WI and 'VQC Pass/Fail Criteria’. B. Most critial points during VQC of pin connectors are: - The pin must be in the middle of the gel, - If metal part of top of the pin on either is not covered with M2 then it is not properly protected. - The M2 must 'locked in place' on all sides and underneath the pin	 

Process Description	M2 Repair	**Health and Safety / PPE**
Element Description	M2 Application on ZIF connector **P2i Confidential - For use by Motorola Repair Centers Only**	Prior to commencement please read relevant risk assessment for Hazards and Risks identified
Operator level required:	Level 2-Intermediate	

Safety key point	Critical point	O	Quality check		
P	Poka Yoke	3	Standard process		Knack point

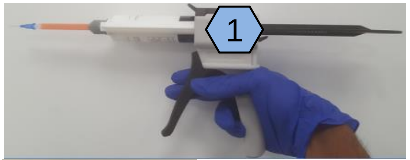
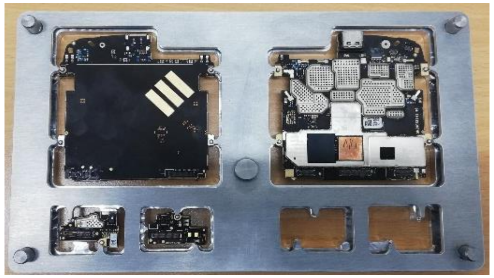
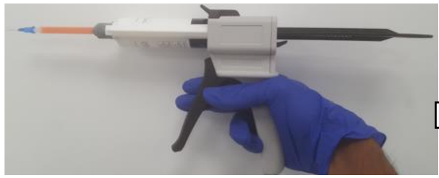
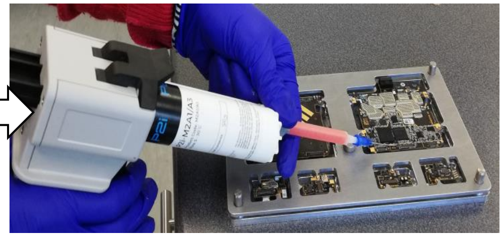
Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

Major Step		Key	Operation detail	Explanation/Examples/Diagrams
1	**Prepare area**		A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. M2 Dispensing Gun 2. Blue Roll 3. Re-work jig with PCBAs **C. Only authorized or trained personnel to undertake this task.**	1  2  3 
2	**Set up**	3	A. Ensure that the dual syringe is clipped into place.Hold the gun with the dominant hand and the other used for support B. Ensure M2 curing check has been performed prior M2 application onto product. B. Dispense some M2 onto a paper towel to remove any air bubbles and ensure the M2 is coming out of the gun properly.	 

3	**Point Specfic Application**	3	A. Ensure ZIF connector is in open position (white ZIF hinge lifted) B. To apply, bring the grey handle towards you to dispense the material, whilst slowly moving the needle tip above the part of the PCBA which is being covered. C. Only one line of M2 is necessary to be applied on the top of black terminal.	 
4	**VQC**	Q	**A. After curing perform VQC as per VQC WI and 'VQC Pass/Fail Criteria'.** B. Most critial points during VQC of B2B connectors are: -M2 should not enter rear side or the ZIF hinge will not clamp -M2 should rest against the white hinge -M2 must not overhang edge of connector - Cannot overhang front of connector or M2 risks catching ZIF flex when inserted	

Process Description	M2 Application Specific					**Health and Safety / PPE**
Element Description	M2 Dispensing		**P2i Confidential - For use by Motorola Repair Centers Only**			Prior to commencement please read relevant risk assessment for Hazards and Risks identified
Operator level required:	Level 2-Intermediate					

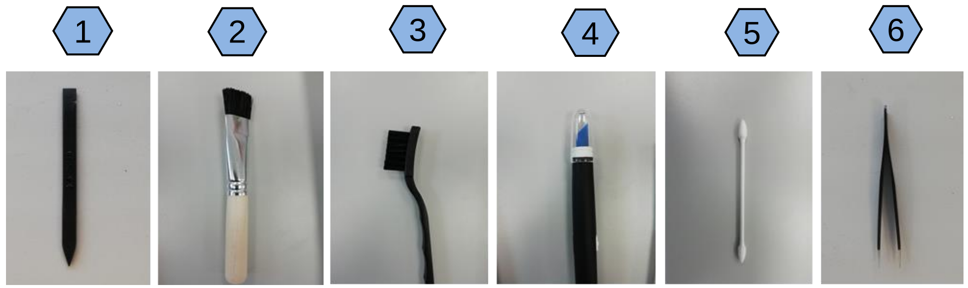
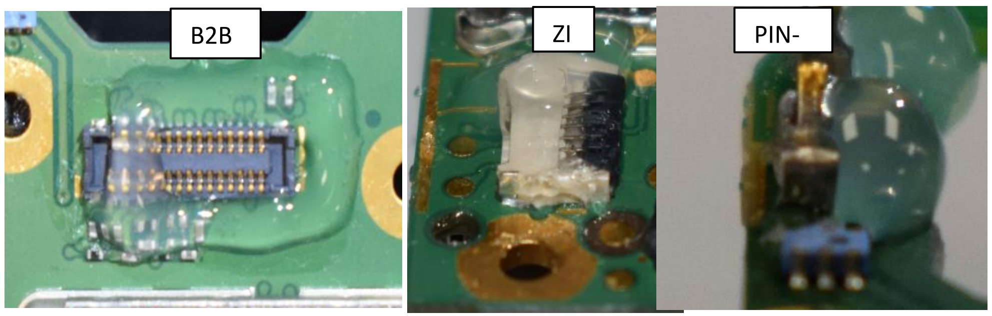
	Safety key point		Critical point	O	Quality check	Safety Shoes		Lab coat	✓
						Light Eye Protection		Hearing Protection	✓
P	Poka Yoke	3	Standard process		Knack point	Goggles			
						Gloves - Polyflex			

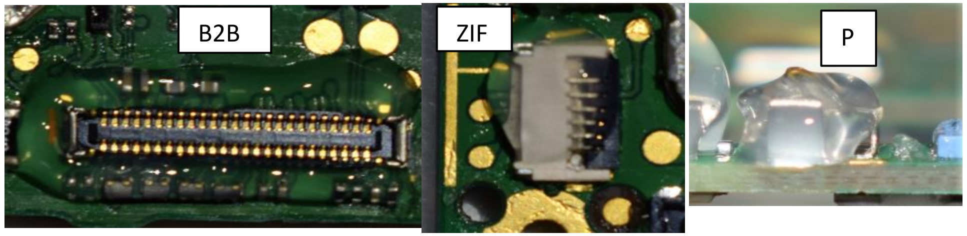
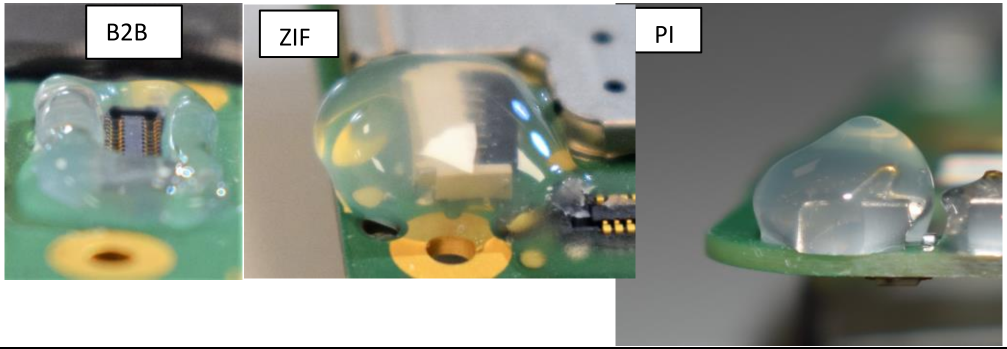
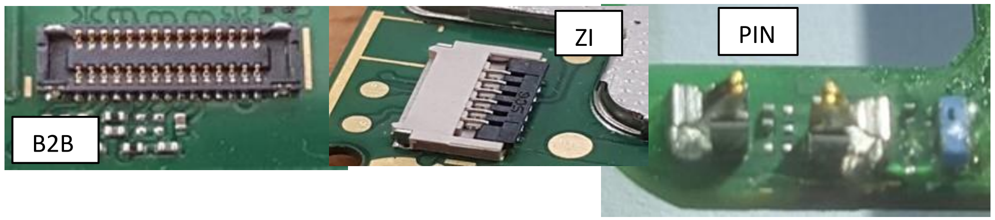
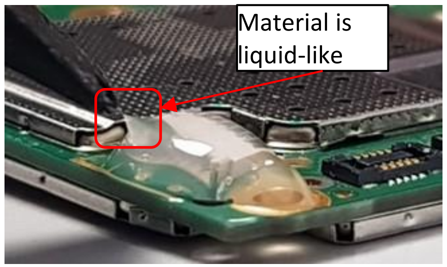
Major Step		Key	Operation detail	Explanation/Examples/Diagrams
1	Prepare area		A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. M2 Dispensing Gun 2. Blue Roll 3. Re-work jig with PCBAs **C. Only authorized or trained personnel to undertake this task.**	1  2  3 
2	**Application**	3	A. Ensure that the dual syringe is clipped into place.Hold the gun with the dominant hand and the other used for support B. Dispense some M2 onto a paper towel to remove any air bubbles and ensure the M2 is coming out of the gun properly. C. To apply the M2, bring the grey handle towards you to dispense the material, as shown in figure 1, whilst slowly moving the needle tip above the part of the PCBA which is being covered.	 

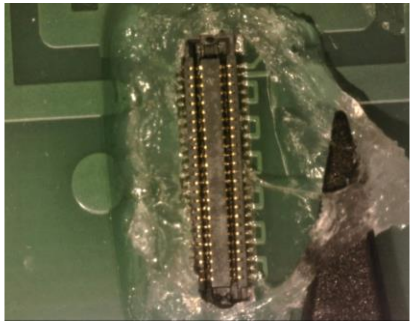
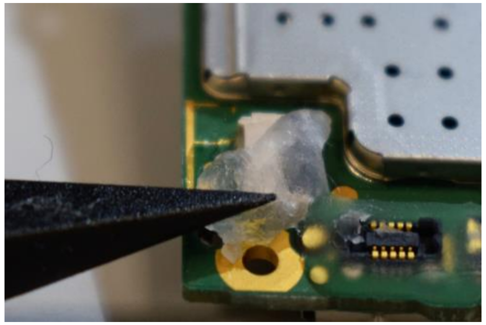
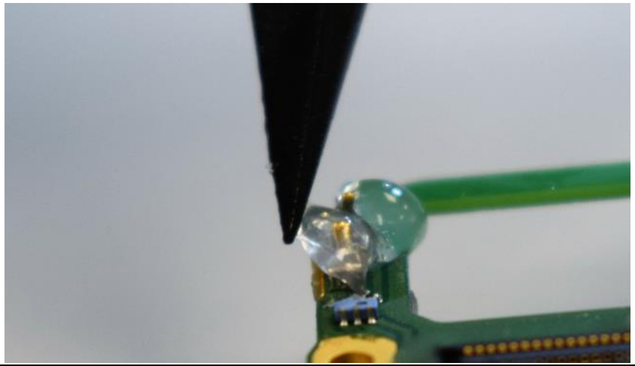
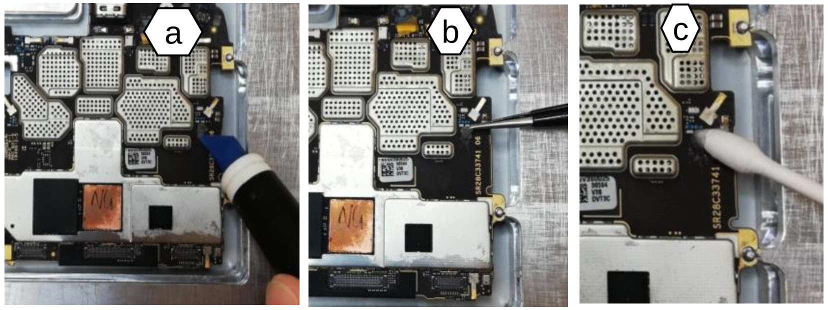
Process Description	M2 Repair	**Health and Safety / PPE**
Element Description	M2 material removal **P2i Confidential - For use by Motorola Repair Centers Only**	Prior to commencement please read relevant risk assessment for Hazards and Risks identified
Operator level required:	Level 2-Intermediate	

	Safety key point		Critical point	O	Quality check
P	Poka Yoke	3	Standard process		Knack point

Safety Shoes		Lab coat	✓
Light Eye Protection		Hearing Protection	✓
Goggles			
Gloves - Polyflex			

Major Step		Key	Operation detail	Explanation/Examples/Diagrams
1	Prepare area		A. Ensure work bench and the surrounding area is tidy and free from obstructions and hazards. **B. Tools and equipment required:** 1. Linear Spudger ESD safe 2. Natural Bristle PCB and Flux Brush 3. Anti Static ESD Cleaning Brush 4. Silicone Cleaning Tool 5. Swab Pointed, Cotton, Double Ended 6. Metal Fine Point Tweezers **C. Ensure ESD strap is connected and worn at all time.** **D. Only authorized or trained personnel to undertake this task.**	
2	**Re-work type:** M2 misplaced	Q	**A. First, make sure M2 is cured. If not, cure it.** B. Approach is M2 removal and re-application.	

<table>
<tr><td>3</td><td>Re-work type: M2 insufficient</td><td>Q</td><td>A. For B2B, ZIF and PIN-type connectors, the appropriate approach is filling up the insufficient gaps with M2.</td><td>
</td></tr>
<tr><td>4</td><td>Re-work type: M2 too much</td><td>Q</td><td>A. For B2B, ZIF and PIN-type connectors, the appropriate approach is M2 removal and re-application.</td><td>
</td></tr>
<tr><td>5</td><td>Re-work type: M2 missing</td><td>Q</td><td>A. For B2B, ZIF and PIN-type connectors, the appropriate approach is M2 application.</td><td>
</td></tr>
<tr><td>6</td><td>Re-work type: M2 uncured</td><td>Q</td><td>A. For B2B, ZIF and PIN-type connectors, the appropriate approach is M2 curing.</td><td>
</td></tr>
</table>

7	Remove M2 from B2B/ZIF connectors		A. Use the spudger to pull M2 off. B. To remove M2 residues: -wet the surface with IPA; - **take care to not damage or scratch plasma coating off nearby components**	 
8	Remove M2 from PIN-type connectors		A. Use a tweezer or a spudger to pull off M2 material. Use IPA if required.	
9	Cosmetic removal of M2 material		A. To remove the cosmetic M2 contamination use: a. Silicone Cleaning Tool b. Metal fine point tweezer/spudger c. Swab pointed cotton **B. Ensure M2 removal is performed very gently without rubbing the surface to avoid damaging the plasma coating layer underneath.**	

10	Pass/Fail criteria for M2 removal		A. Ensure that M2 material has been completely removed, before passing the PCBA board to the next VQC stage.	PASS FAIL FAIL